

Application of the Linear Partial Information Model to Forecasting the Swiss Timber Market

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ABSTRACT

Most often, statistical analyses only provide partial information about the appropriateness of different models, structures and parameters which may underlie the dynamic process that has generated a time series. Linear partial information (LPI), in particular, consists of linear restrictions such as LPI: $p_a > p_b$, $p_a > p_c$ where p_a denotes the probability that structure a holds. Fuzzy information of this type can be put to use for decision-making by LPI analysis. In this paper, LPI analysis is applied to answer the question of whether subsidizing price, given an abnormal disturbance on the timber market, would contribute to continuous forest management, a stated goal of Swiss environmental policy.

KEY WORDS Linear partial information Games against nature
Fuzzy information Model uncertainty
MaxEmin principle

INTRODUCTION

Forecasting based on economic theory encounters at least three major problems. First, the structure of the postulated optimization processes of individual economic agents is only partially reflected by aggregate economic time series. Secondly, although there exist theoretical restrictions as well as statistical findings providing some information about interactions of variables, statistical tests rarely discriminate between different structures of a given model. Finally, future realizations of the variable to be forecast continue to be subject to random disturbances.

These three problems will be addressed in this paper. The tool used is LPI analysis, which systematically exploits linear partial information. For example, although it may be impossible to ascertain that structure (parameter set) A of a given model is the true one, it may still be possible to state that structure A is more likely to be true than either B or C. More generally, LPI analysis deals with restrictions in the guise of systems of linear inequalities defined over probabilities.

In this paper, LPI analysis is applied to the Swiss timber market. The so-called *Waldsterben* ('death of forests') is the primary source of concern for the future development of Swiss forestry. Increased timber harvests in Switzerland and in Europe may (as a result of government

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interventions) render previous modelling efforts obsolete or cause changes of parameters (through changes in timber stocks). The Swiss Parliament, in its session of February 1985, has discussed possible policies for the stabilization of prices of Swiss timber to ensure continuity of forest management (Schärer and Zimmermann, 1984). In this paper, we focus on predicting Swiss timber prices. If the price is high, incentives for tree growers to remain in business are sufficiently strong. If it is low, a subsidy may be necessary.

This paper is organized as follows: in a first step, the decision problem facing the Swiss Parliament is defined. This analysis reveals that uncertainty regarding the appropriate model, appropriate structure and choice of disturbance are important aspects of the problem. Next, results of time series analysis are exploited for providing LPI restrictions concerning the credibility of three different structures. An LPI analysis of the partial information available leads to a prediction of Nature's optimal choice. Another section deals with the problem of fuzzy information as to the likelihood of normal or abnormal disturbances. In that context, the possibility of subsidies inducing abnormal disturbances is taken into account. Finally, Nature's choice among models is determined using LPI methods once more. The final section contains a discussion of the optimal decision to be taken by the Swiss Parliament and some of the open questions that should be tackled in future research.

THE DECISION PROBLEM DEFINED

In this section, the decision problem of the Swiss Federal Parliament in view of a possible subsidy of forestry is outlined. The Swiss Parliament, as well as the *Bundesrat* (federal executive council) have signalled concern for Swiss forests because a considerable reduction of woodland would increase environmental dangers, such as avalanches in the Alps and flood disasters. It may also cause loss of employment in the wood processing and forest industries as well as in tourism. The goal of stable forest acreage can only be achieved if earnings in forestry are sufficient to keep employment at about present levels. Since there are no price regulations, a rise in the supply of timber will, *ceteris paribus*, reduce the price of timber, jeopardizing continuous management of forests in Switzerland. Forecasts of Swiss timber prices therefore provide the Swiss Parliament with important information concerning the need for granting subsidies to the timber business.

The prediction of future developments is a classical application of econometric analysis. Traditional econometric forecasts are based on a single equation for one dependent variable, pretending that certain parameters of dynamic optimization by individual agents (such as the planning horizon) are known (Sims, 1980). This assumption is invalidated by the observation that competing econometric models coexist for the same market and the same period of time. Even worse, different structures (sets of parameter estimates) for the same model coexist, depending on the period of observation selected for statistical analysis. Linear partial information (LPI) analysis of competing structures provides a systematic method for exploiting the fuzzy information about the forecasting performance of an econometric model at hand.

Figure 1 contains a decision tree that serves to illustrate the decision-making situation. Node D_0 , under the control of the decision-maker, is marked as a box; nodes under the control of Nature are marked as circles. At node D_0 , Parliament has to decide whether a special subsidy should be allocated to the Swiss forest industry. This decision will mainly be based on a forecast of selling prices received by Swiss tree growers (see the right margin of Figure 1). The upper half of the decision tree applies if a subsidy is granted by Parliament. Rough estimates indicate that a subsidy would amount to 3.00 Swiss francs (SFr.) per cubic metre (m^3). Uncertainty about

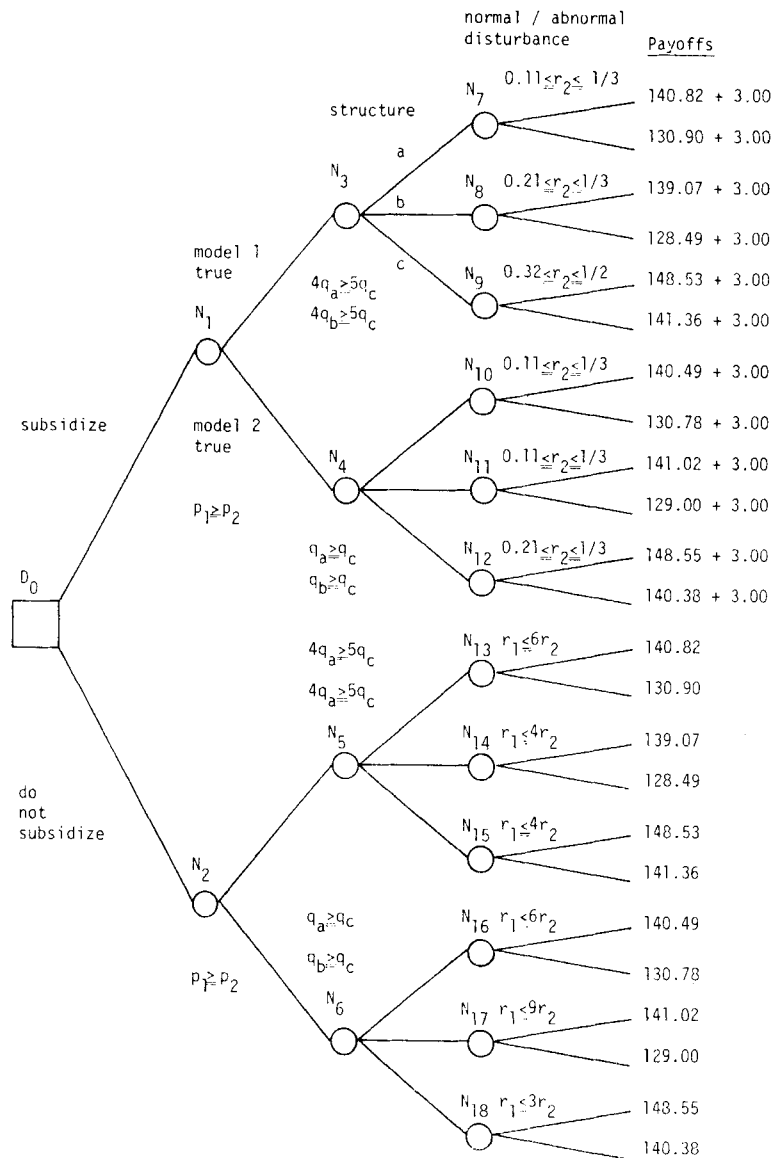


Figure 1. Decision tree: the Swiss Parliament and the timber market

this figure could be easily taken into account using LPI analysis in a way analogous to that developed below. Admittedly, this is only an intermediate payoff; but high timber prices serve as an incentive for cultivating wood acreage.

The predicted timber price depends on the model used; for example, given normal circumstances, the expected net price is 139.07 SFr./m³ (node N_8) according to model 1, whereas it amounts to 141.02 SFr./m³ (node N_{11}) according to model 2. At nodes N_1 and N_2 , Nature decides whether model 1 or model 2 prevails. All that can be distilled from an analysis of the historical forecasting performance of the two competing models is linear partial information (LPI). Here it takes the

form LPI: $p_1 \geq p_2$, i.e. model 1 is more likely to be the true one than model 2. Derivation of this LPI will be the subject of a later section. But even if the true model were known to the decision-maker, this is not sufficient for calculating a price forecast. For that, it takes a set of estimated parameter values, i.e. a structure.

Unfortunately, there are several plausible sets of such parameters. For the estimated coefficients fail to be stable; rather, they heavily depend on the period of observation selected. There are many reasons for such instability (Menges and Skala, 1974), but in this context reference should be made to export subsidies in 1967/68, changing regulations concerning forest management, and new means of transport for timber (Meyer, 1979). In view of this instability, the choice of structure may become as important as the choice of the model itself. On the basis of the forecasting record examined below, there are three competing sets of parameters, with Nature selecting one of them as correct (node N_3). Their credibility can be expressed by the LPI: $4q_a \geq 5q_c$, $4q_b \geq 5q_c$, where q_a denotes the probability of structure a being the true one (see next section). This restriction might in turn be qualified by a competing restriction, with an LPI expressing uncertainty as to their respective validity (Kofler and Zweifel, 1984).

Nature has a last say at nodes N_7 to N_9 , e.g. when it comes to taking into account the impact of special events. Depending on the model retained (1 or 2) and the structure used (parameter sets a , b or c) there are different LPIs concerning the probability of abnormal disturbances. Whereas prices net of subsidy are the same regardless of the decision taken by Parliament, the probability that 'abnormal disturbances' occur is higher as soon as a subsidy is at stake. For this reason, the LPI at node N_7 , for instance, differs from that of node N_{13} , a complication that will be dealt with below.

DERIVING LPI RESTRICTIONS

In this section, time series analysis is used as a basis for deriving LPI restrictions of the type entered in the decision tree of Figure 1. In the context of estimating vector autoregressive model No. 1, it turned out that parameter values depended very much on the period selected. This is illustrated for model 1 (Schelbert, 1985), which reads

$$Y = A(L)Y + B(L)X_1 + C(L)X_2 + e_1 \quad (1)$$

with Y the Swiss timber price, X_1 the Austrian board price and X_2 the German board price, all in Swiss francs per m^3 .

Optimal lag structures for $A(L)$, $B(L)$ and $C(L)$ were obtained using the final prediction error criterion (Akaike, 1969). This model was estimated for periods I, II and III as defined in Table 1, resulting in structures a , b and c .

Root mean square errors were calculated, yielding almost identical values for structures a and b (see Table 2). In this situation, no statement as to the relative validity of structures a and b can be made. However, structure c is clearly dominated by structures a and b .

Similar results were found for model 2, where again structure c is dominated by both a and b , with no clear difference between a and b . Therefore, in the case of model 2 the natural LPI is $q_a \geq q_c$, $q_b \geq q_c$. But when the two models are compared, it becomes clear that the degree of dominance over structure c is much greater in model 1 than model 2. As a matter of fact, structure c in model 1 has a root mean square error which is 1.5 times that of structure a , whereas this ratio is only 1.17 in model 2. Thus, in a certain sense structures a and b are 'more dominant' in model 1 than in model 2, the factor amounting to about $1.25 = 5/4$ (see last column

Table 1. Characterization of subperiods distinguished in estimation

Period	Characterization	Abnormal disturbances
Period I, 50/I–59/III	No abnormal disturbances	no
Period II, 60/I–70/III	Winter 1961/1962: extensive storm damages, but extra supply of timber coincides with strong demand.	no
	February/March 1967: storm damage on a national scale, major price drop: export subsidy of 8 Mio. SFr. allocated.	yes
Period III, 71/I–81/III	1975: storm damages; extra supply coincides with shrinking demand.	yes
	1982: Shinook storms; major price drop.	yes
82/I–84/III	Test of forecasting performance: Model 1 LPI: $4q_a \geq 5q_c, 4q_b \geq 5q_c$ Model 2 LPI: $q_a \geq q_c, q_b \geq q_c$	

Structure *a* is based on periods I, II and III; structure *b*, on II and III; structure *c*, on III only. Time units consist of 4 months; e.g. 50/I denotes January to April, 1950.

Source: Forest Statistics, 1950/I–1984 III, BIGA, Bern; and Jakob (1985).

Table 2. Deriving LPI restrictions from root mean square errors (RMSE)

	Structure			Ratio, a/c	$\frac{a/c \text{ model 1}}{a/c \text{ model 2}}$
	a	b	c		
	RMSE				
Model 1	5.47	5.23	8.16	1.5	1.25 = 5/4
LPI 1:	$4q_a \geq 5q_c, 4q_b \geq 5q_c$				
Model 2	6.09	6.19	7.12	1.17	
LPI 2:	$q_a \geq q_c, q_b \geq q_c$				

q_a = probability of structure *a* being the true one; $q_a + q_b + q_c = 1$.

of Table 2). Thus, the LPI entered at node N_3 reads LPI: $q_a \geq (5/4)q_c$ or $4q_a \geq 5q_c$; $q_b \geq (5/4)q_c$ or $4q_b \geq 5q_c$ (see node N_3 in Figure 1).

CHOICE OF STRUCTURE USING LPI ANALYSIS

This section is devoted to the problem of choosing among competing parameter estimates (structures) for a given model when partial information in the guise of LPI (as derived above) is available. The issues related to abnormal disturbances and model selection will be dealt with in the following sections. In order to illustrate the use of LPI analysis, we focus on stochastic node N_3 , taking expected payoffs at subsequent stochastic nodes N_7 to N_9 as given (these payoffs will also be determined later on).

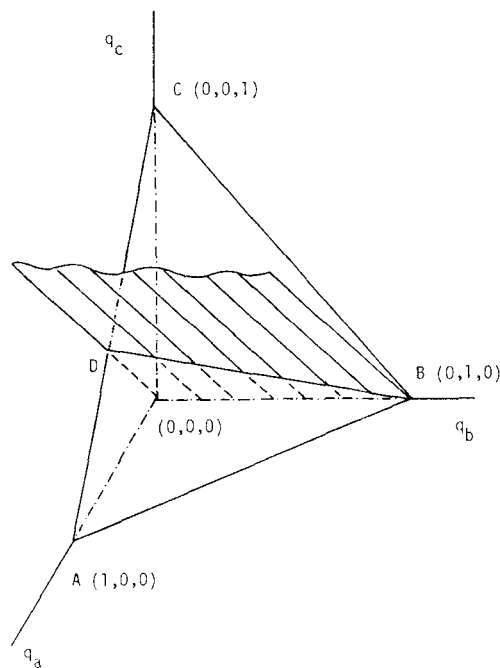


Figure 2. LPI in three dimensions

In principle, probabilities (q_a, q_b, q_c) associated with the three structures may vary freely between zero and one, subject only to the restriction $q_a + q_b + q_c = 1$. Graphically, the set of admissible values of (q_a, q_b, q_c) is given by the triangular plane ABC shown in Figure 2. However, there are additional LPI restrictions, derived in the preceding section. In Figure 2, only the restriction LPI: $4q_a \geq 5q_c$ ($4q_a = 5q_c$ in the limit) is entered. It reduces the admissible set to the triangular plane ADB of Figure 2.

Transformed into two rather than three-dimensional space, the set of distributions satisfying the summation condition is given by the set of points contained in the barycentric triangle of Figure 3. The restriction LPI: $4q_a \geq 5q_c$ will be recognized as the straight line DB. Figure 3 also contains the second restriction LPI: $4q_b \geq 5q_c$. In all, the LPI reduces the set of admissible values

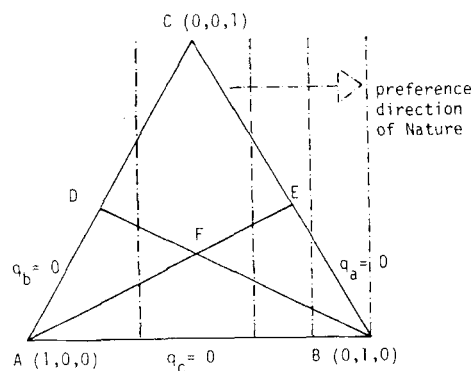


Figure 3. LPI in two dimensions

of (q_a, q_b, q_c) , i.e. the set of admissible probability distributions defined over the three structures, to the small triangle AFB shown in Figure 3.

This reduced set is still not finite, however. In order to make the set of points (distributions) to be examined finite, a crucial assumption is needed. This assumption states that the decision-maker is engaged in a game against Nature. Since co-operation with Nature can be excluded, the game is zero-sum (or more generally, constant sum) in expected payoffs (Wald, 1950; Zweifel, 1982). This implies that Nature is an antagonistic player with regard to the one objective defined here; in the example, she presumably pulls towards vertex B because the expected payoff is maximal for her and minimal for the decision-maker. This gives rise to a preference field characterized by the straight indifference lines appearing in Figure 3. More generally, the assumption introduced above allows us to conclude that Nature will select one of the three vertices of the admissible set. However, the number of such vertices is finite; for more details regarding the axiomatic foundations of LPI analysis, the reader is referred to Kofler (1985), Kofler *et al.* (1984) and Kofler and Zweifel (1984). An appendix containing the relevant theorems is available upon request from the authors.

A basic theorem of linear programming says that at such a vertex, the number of restrictions satisfied as exact equalities is equal to the number of variables involved (Intriligator, 1971, p. 490). This theorem can be directly applied here to derive the extremal points of Figure 3, or equivalently the extremal, decision-relevant distributions (f_1 to f_3):

LPI _{N3} :	f_1	f_2	f_3
$4q_a \geq 5q_c$	×		×
$4q_b \geq 5q_c$	×	×	
$q_c \geq 0$		×	×
$q_a + q_b + q_c = 1$	×	×	×

 $\rightarrow \mathbf{M}(\text{LPI})_{N3} = \begin{bmatrix} f_1 & f_2 & f_3 \\ 5/14 & 1 & 0 \\ 5/14 & 0 & 1 \\ 4/14 & 0 & 0 \end{bmatrix}$

Here, restrictions satisfied as exact equalities are marked. In the first column of $\mathbf{M}(\text{LPI})_{N3}$, for instance, the first two restrictions are satisfied as equalities along with the summation restriction. Thus, for three variables, exactly three restrictions are satisfied, in accordance with Intriligator (1971, p. 490).

Given these extremal distributions, it is an easy task to calculate the expected values, using the payoffs given in Figure 1, nodes N_7 to N_9 :

$$\begin{aligned} \mathbf{E}(u)_{N3} &= (140.51 \quad 138.54 \quad 147.95) \begin{bmatrix} 5/14 & 1 & 0 \\ 5/14 & 0 & 1 \\ 4/14 & 0 & 0 \end{bmatrix} \\ &= (141.93 \quad 140.51 \quad 138.54) \end{aligned}$$

Depending on the distribution over the three structures (a, b, c) selected by Nature, expected payoff varies between 141.93 and 138.54. Nature, according to the basic game-theoretic assumption (minEmax criterion; see Kofler *et al.*, 1984), will choose that distribution which yields minimal expected payoff to the decision-maker, i.e. f_3 with 138.54 (see also the preference field pointing to point B (0, 1, 0) in Figure 3). Thus, the value of the decision-maker's game against Nature at node N_3 is 138.54. Applying the same procedure to nodes N_{10} to N_{12} , using the LPI relevant at node N_4 , yields the following expected payoffs:

LPI _{N4}	f_1	f_2	f_3
$q_a \geq q_c$	×		×
$q_b \geq q_c$	×	×	
$q_c \geq 0$		×	×
$q_a + q_b + q_c = 1$	×	×	×

 $\rightarrow \mathbf{M}(\text{LPI})_{N4} = \begin{bmatrix} f_1 & f_2 & f_3 \\ 1/3 & 1 & 0 \\ 1/3 & 0 & 1 \\ 1/3 & 0 & 0 \end{bmatrix}$

$$\begin{aligned}
 E(u)_{N4} &= (140.25 \quad 140.01 \quad 148.83) \begin{bmatrix} 1/3 & 1 & 0 \\ 1/3 & 0 & 1 \\ 1/3 & 0 & 0 \end{bmatrix} \\
 &= (143.03 \quad 140.25 \quad 140.01)
 \end{aligned}$$

Again, among these values only the minimal expected payoff, amounting to 140.01, is relevant. According to the minEmax criterion, Nature will select distribution f_3 (0, 1, 0).

THE ROLE OF ABNORMAL DISTURBANCES

A glance at the payoff column of Figure 1 reveals that predicted price varies a great deal depending on whether or not the year ahead is thought to be a normal one. Deciding whether a disturbance is exceptional or not requires refined analysis, because the histogram in Figure 4 clearly shows that the classical assumption of a normal distribution cannot be maintained. It cannot simply be replaced by another assumed distribution either, for histograms take on quite different shapes at other nodes. In this situation, calculating confidence intervals and declaring a residual as being beyond confidence limits with an exact probability would be haphazard (Doan *et al.*, 1984).

However, the histogram shown in Figure 4 still provides some partial information. In particular, as long as second moments exist, the standard error s of the residual may be used to create a $(-\hat{s}, \hat{s})$ interval. The mass lying below the $(-\hat{s})$ threshold was calculated and compared

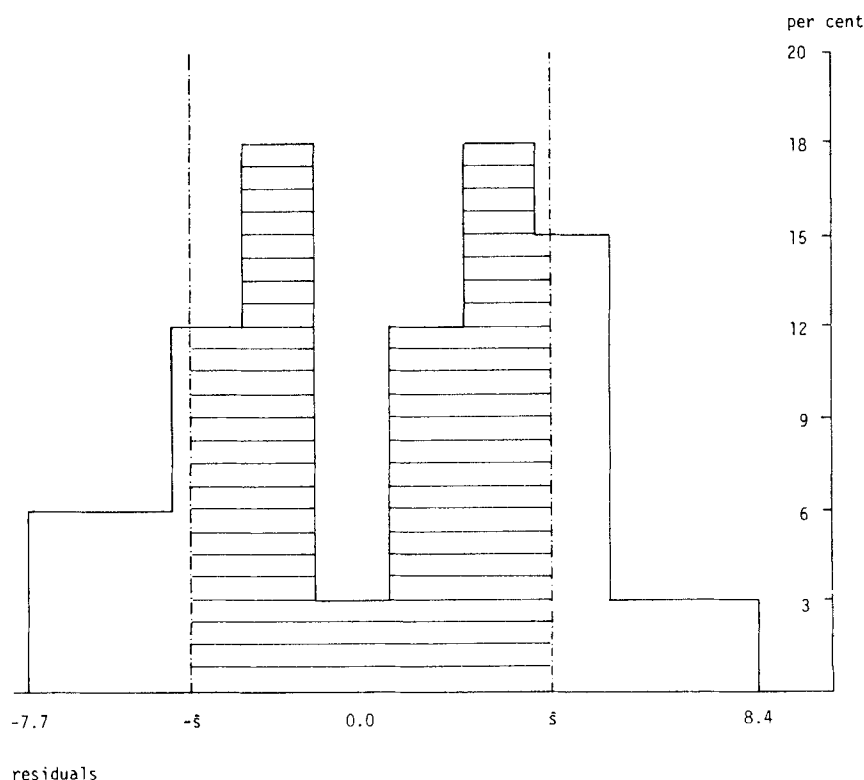


Figure 4. Distribution of residuals for model 1, structure c

to the rest. For example, there is an estimated likelihood of no more than 0.14 for the residuals for model 1 (structure *a*) to lie below the $(-\hat{s})$ limit. Exceptionally high positive residuals are of little interest because they will never call for action by political authorities. Only exceptionally low timber prices may give rise to a subsidy and vice versa, a complication that will be taken into account below.

Now the information that a lower bound $(-\hat{s})$ was exceeded with a likelihood of no more than 14 per cent can be exploited to yield the LPI: $r_1 \geq 6r_2$ where r_1 denotes the probability of a normal disturbance occurring (see node N_{13} of Figure 1). A critic may note that an LPI: $r_1 \geq 2r_2$ would have been acceptable as well. It would be an easy task to solve the decision tree using this LPI' and to show the loss of value associated with this weakening of the LPI restriction (Kofler, 1985). LPIs for the remaining lower half of the decision tree can be derived in analogous manner.

As to the upper half of the decision tree, previously granted subsidies (see Table 1) may have initiated learning processes. Thus it is conceivable that the likelihood of an abnormal disturbance in timber price increase if Parliament decides in favour of a subsidy; for instance a winter storm of only local importance may furnish the pretext for getting rid of a great deal of substandard timber at the subsidized price. This complication is dealt with in the following way. Directly estimating the probability $P(D|S)$, i.e. the probability of an abnormal disturbance occurring given that the subsidy has been granted, is not possible because subsidies have been too rare an event. Instead, an indirect estimate is derived using the Bayes formula:

$$P(D|S) = P(S, D)/P(S)$$

$P(D|S)$ the probability of an abnormal disturbance occurring given the subsidy, $P(D, S)$ the probability of the abnormal disturbance and the subsidy occurring jointly and $P(S)$ the probability of a subsidy being allocated.

Once more, there are different estimates of both the numerator and the denominator. As to the numerator, instead of the $(-\hat{s}, \hat{s})$ interval, one might have used a $(-2\hat{s}, 2\hat{s})$ interval for determining abnormal disturbances. For example, at node N_7 (model 1, structure *a*) there are two joint observations of D and S using the first interval, whereas there is only one using the second. Since there are 96 observations, we obtain the LPI: $1/96 \leq P(S, D) \leq 2/96$.

As to the denominator, the probability estimate of $P(S)$ depends very much on the observation period selected. Overall, there have been six instances in 96 observations of a subsidy being granted. However, these grants occurred within the subperiod associated with structure *b*, containing 61 observations associated with structure *b* (see Table 1). Thus, there is a natural restriction on $P(S)$, namely LPI: $6/96 \leq P(S) \leq 6/61$.

These two restrictions jointly give rise to an LPI concerning $P(D|S) = r_2$ in the upper part of the decision tree of Figure 1. The bounds are derived as follows:

$$\begin{aligned} \min P(D|S) &= \frac{\min P(D, S)}{\max P(S)} \\ \text{for node } N_7: \min r_2 &= \frac{(1/96)}{(6/61)} = 0.11 \\ \max P(D|S) &= \frac{\max P(D, S)}{\min P(S)} \\ \text{for node } N_7: \max r_2 &= \frac{(2/96)}{(6/96)} = 1/3 \end{aligned}$$

Table 3. minEmax values of the game against Nature of Figure 1

Decision node D_0	Model nodes	Structure nodes	Disturbance nodes
$D_0 = 138.54$	$N_1 = 138.54$	$N_3 = 138.54$	$N_7 = 140.51$
			$N_8 = 138.54$
			$N_9 = 147.95$
	$N_2 = 136.95$	$N_4 = 140.01$	$N_{10} = 140.25$
			$N_{11} = 140.01$
			$N_{12} = 148.83$
	$N_2 = 136.95$	$N_5 = 136.95$	$N_{13} = 139.45$
			$N_{14} = 136.95$
			$N_{15} = 146.14$
	$N_2 = 136.95$	$N_6 = 139.10$	$N_{16} = 139.10$
			$N_{17} = 138.82$
			$N_{18} = 146.51$

Putting these bounds for node N_7 together, we obtain the LPI: $0.11 \leq r_2 \leq 1/3$. This is a sample of an LPI generalization of Bayesian inference. It is theoretically justified by Kofler and Menges (1976, pp. 120–121) and by Kofler and Zimmermann (1985, chap. 2); for another, somewhat more complicated, example see Kofler and Zweifel (1984). The LPIs appearing at nodes N_8 to N_{12} of Figure 1 are derived in an exactly analogous manner. There is merely a simplification in that the quantity $P(D, S)$ is invariant to the choice of interval at nodes N_8 , N_9 and N_{12} ; in these instances, extending the interval for ‘abnormal’ disturbances to $(-2\hat{s}, 2\hat{s})$ does not matter. The resulting restrictions are entered in the upper half of the decision tree. Given these LPI, the payoffs at nodes N_7 to N_9 that were used in the previous section can be derived at once. This is illustrated for node N_7 :

$$\begin{aligned} E(u)_{N_7} &= (143.82 \quad 133.90) \begin{bmatrix} 0.89 & 2/3 \\ 0.11 & 1/3 \end{bmatrix} \\ &= (142.72 \quad 140.51) \end{aligned}$$

Since Nature is assumed to be an antagonistic player, only the minimum expected payoff is to be taken into account in the subsequent evaluation of node N_3 . A general overview of decision-relevant payoffs is given in Table 3.

EVALUATING COMPETING MODELS WITH LPI

In terms of goodness of fit as well as forecasting performance, model 1 is not clearly superior to another specification (model 2):

$$Y = A(L)Y + D(L)X_1 + F(L)X_3 + e_2 \quad (2)$$

with X_3 the Italian timber price, in Swiss francs per m^3 .

Since prediction errors under all three structures (a, b, c) are somewhat higher than they are in model 1 (see Table 2), there is some justification for postulating

$$\text{LPI: } p_1 \geq p_2$$

with p_1 the probability of model 1 being the true one.

This LPI gives rise to the simple extremal matrix used below for the evaluation of N_1 :

$$\begin{aligned} E(u)_{N_1} &= (138.54 \quad 140.01) \begin{bmatrix} 1 & 1/2 \\ 0 & 1/2 \end{bmatrix} \\ &= (138.54 \quad 139.28) \end{aligned}$$

Proceeding in an exactly analogous manner, the node N_2 (the lower branch of the decision tree in Figure 1) can be evaluated:

$$\begin{aligned} E(u)_{N_2} &= (136.95 \quad 139.10) \begin{bmatrix} 1 & 1/2 \\ 0 & 1/2 \end{bmatrix} \\ &= (136.95 \quad 138.03) \end{aligned}$$

CONCLUSION AND OUTLOOK

We now finally turn to the decision-maker. Being at decision node D_0 of Figure 1, he will simply compare the value of the game at nodes N_1 and N_2 . At node N_1 (allocate a subsidy), the guaranteed expected value of the game is 138.54. At node N_2 (do not subsidize), the guaranteed expected value of the game amounts to 136.95. Applying the maxEmin criterion (maximizing minimal expected payoff) once more, we conclude that a subsidy should be granted—a clear-cut suggestion for policy in spite of the fuzziness of the data on which it is based.

However, two reservations should be made at once. One is that the difference in payoffs is less than three per cent, which means that an alteration of the elements in the decision problem could easily overthrow the suggestion formulated above. Secondly, the difference in expected guaranteed payoffs in terms of sales price for timber is 2.4 SFr. per m^3 whereas the subsidy would amount to 3.00 SFr. per m^3 . Thus there seem to be decreasing marginal returns to the subsidy.

There are at least two ways in which the above analysis could be improved. First, payoffs are not as well determined as they appear. In particular, a given total subsidy could easily result in a smaller amount per m^3 of timber sold, if indeed more timber is put on the market under the pretext of an abnormal disturbance. Thus, it may be quite wrong to suppose that net price will be increased by a fixed amount under all circumstances. Partial information regarding likely impacts of other circumstances on the subsidy received per m^3 of timber could be incorporated in the LPI analysis, however. A second and final point concerns the definition of payoff. Price certainly is an incentive for entrepreneurs to remain in the timber business. Nevertheless, Parliament would rather be interested in seeing wood acreage on the increase or stabilized at prevailing levels. Economic and econometric research efforts can do no more than derive an approximate relationship between timber price received and wood acreage. But it seems clear that the resulting statements would take the form of fuzzy information—information that could be evaluated using further LPI analysis.

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